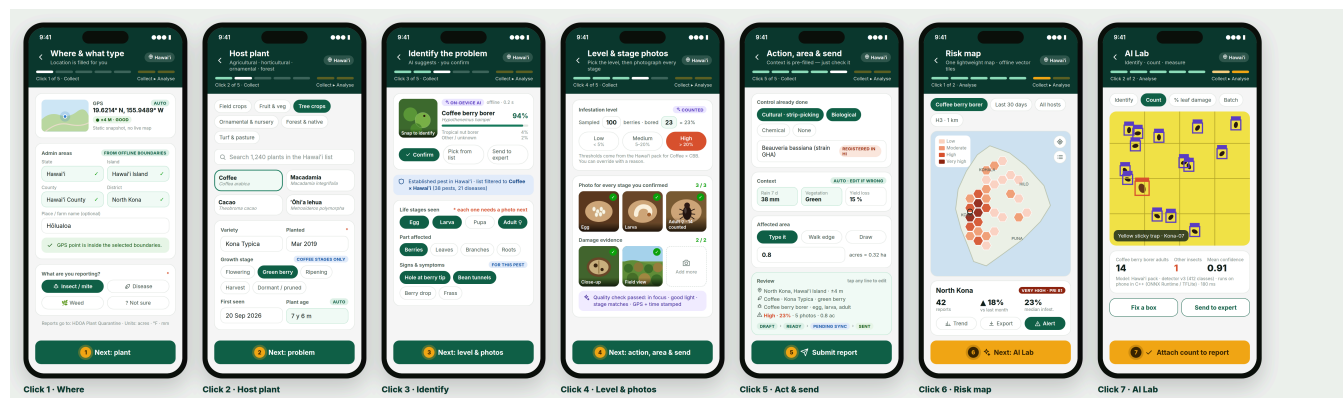


PestLand — User Manual

Plant health surveillance for any plant, any region Version 1.0 · September 2026



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1. What PestLand is

PestLand is a mobile app for reporting and mapping insect pests, plant diseases and weeds on **any plant**: field crops, fruit and vegetables, tree crops, ornamentals and nursery stock, forest and native plants, turf and pasture.

It keeps the backbone of **CropProtect** (where → plant → problem → evidence → context → area → submit) and adds what CropProtect was missing:

Borrowed from	What PestLand adds
ODK Collect / Central	Draft > Ready > Pending > Sent states, validation rules, GPS accuracy check, form versioning, safe re-sync, web review queue
Plantix	Photo-first AI identification, picture cards for plants and pests, photo coaching, many languages
FAO FAMEWS	Standard sampling (count → % → level), trap counts, risk/hotspot maps, early-warning alerts
New in PestLand	Region packs, one photo for every life stage, on-device AI counting, hexagon risk index, only one lightweight map

What PestLand takes from each app

CropProtect's surveillance backbone, plus the missing features from ODK Collect, Plantix and FAMEWS.

From CropProtect (kept)

- Crop-first logic, crop-filtered pest lists
- Pest/disease vs weed branch
- Close-up + field photos
- Rainfall, vegetation, yield loss
- Manual or mapped area
- Offline pending queue

From ODK Collect / Central

- Draft > Ready > Pending > Sent states
- Constraints, skip logic, calculated fields
- GPS accuracy threshold
- Form & pack versioning, audit log
- Encrypted, idempotent sync
- Review/approve queue on the web

From Plantix

- Photo-first AI identification
- Visual plant & pest cards
- Photo quality coaching
- Multi-language UI
- Weather context
- Approved guidance links (not prescriptions)

From FAO FAMEWS

- Standard scouting/sampling protocol
- Counts → % infestation → level
- Trap monitoring
- Risk / hotspot maps
- Early-warning alerts
- Shared, comparable data across regions

New in PestLand

- Region packs (Hawaii, Guam, AU, PNG, US, KE)
- Any plant: farm, garden, ornamental, forest, native
- Photo for every life stage
- On-device C++ AI counting
- H3 hex risk index
- One lightweight map only

Two design rules

- **Simple:** 5 screens to collect a report, 2 screens to see results. Anything the phone can work out (location names, plant age, weather, level from a count) is filled in for you.
- **Light:** CropProtect loaded a Google/Leaflet map on several pages. PestLand opens a map **only** when you draw an area or look at the risk map, and that map uses small offline vector tiles. Click 1 shows a static snapshot, not a live map.

2. Getting started

2.1 Install and allow permissions

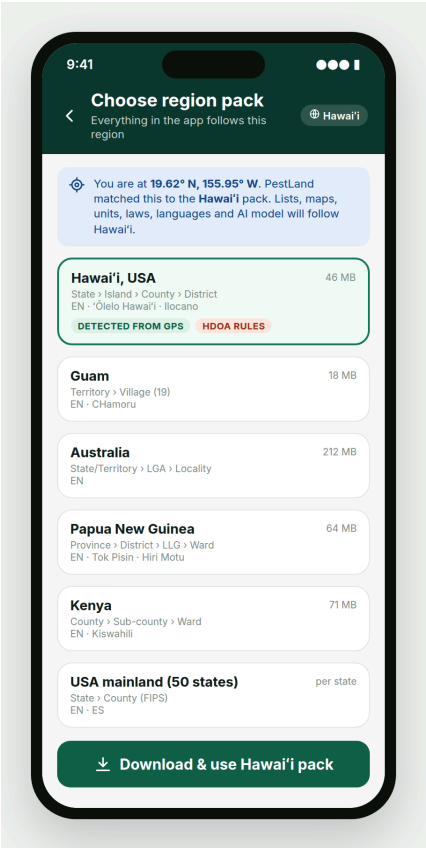
Install PestLand and tap **Allow while using the app** for:

Permission	Why
Location	Fills in GPS and all admin areas in Click 1
Camera	Stage photos, damage photos, AI identification
Storage / photos	Keeps reports and photos offline until they are sent
Notifications (optional)	Regional alerts and "needs your attention" messages

2.2 Sign in

Sign in with your email (or the phone number your region uses). Your role (collector, expert, reviewer or region admin) decides what you see. Collectors see the 5 + 2 clicks; experts also see the ID queue; admins can edit region packs.

2.3 Choose your region pack

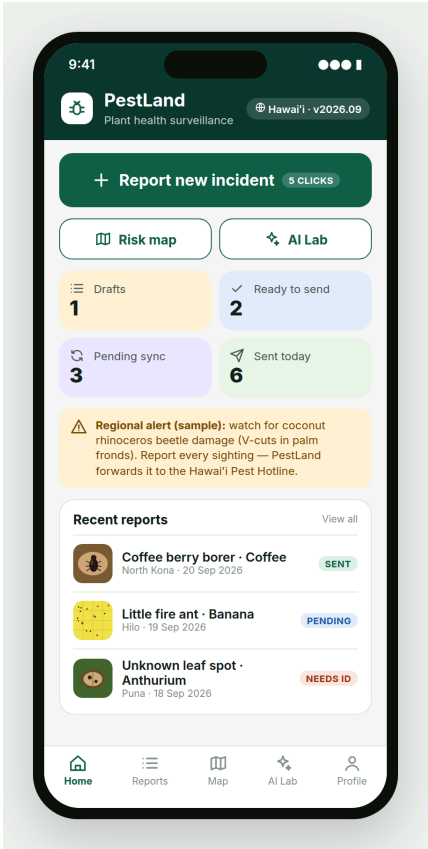


On first launch PestLand reads your GPS and suggests a **region pack**, for example *Hawai'i*. Tap **Download & use**.

A region pack decides **everything** that follows: admin areas, plant list, pest list, legal status of each pest, severity rules, control products, AI model, offline map, language and units. When you pick Hawai'i, every list and map stays inside Hawai'i. See [Section 13](#).

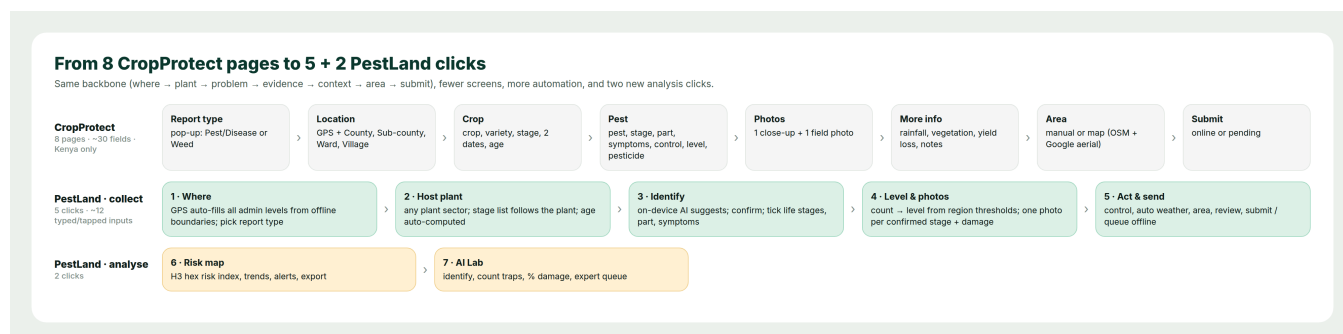
You can hold several packs and switch any time from **Profile > Region**. If you travel to a place your pack doesn't cover, PestLand warns you and offers the matching pack.

3. Home screen



Element	What it does
Report new incident	Starts the 5-click report
Risk map / AI Lab	Jump straight to Click 6 or Click 7
Drafts	Reports you started but have not finished
Ready to send	Finished reports waiting for you to tap Submit
Pending sync	Submitted with no signal. They go automatically when you are back online
Sent today	Reports the server has received
Regional alert	Messages from the region's plant-protection agency
Recent reports	Status tags: <i>Sent</i> , <i>Pending</i> , <i>Needs ID</i> (an expert still has to name the problem)

4. The 5 + 2 clicks at a glance

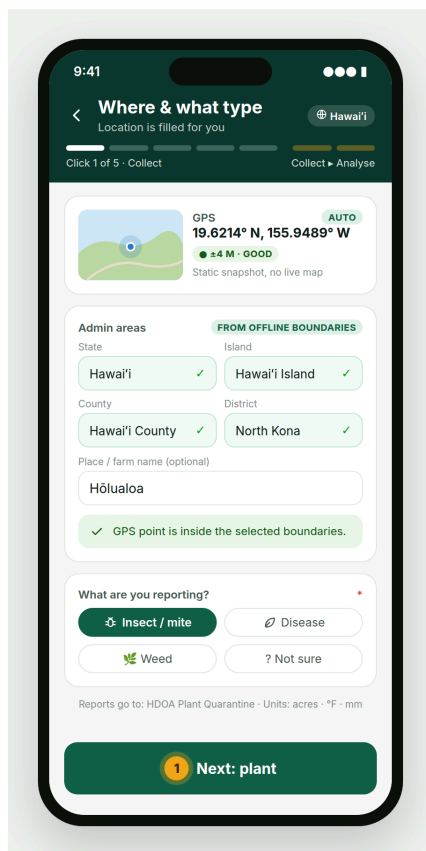


A **click** here means one screen, finished with one **Next** tap. The yellow numbered dot on each Next button shows which click you are on, and the bar at the top shows your progress: 5 green segments for collecting, then 2 amber ones for analysing.

Click	Screen	You do	PestLand does for you
1	Where & what type	Check location, pick Insect / Disease / Weed / Not sure	GPS, accuracy, every admin level, boundary check
2	Host plant	Pick sector and plant, stage, planting date	Stage list for that plant, plant age
3	Identify	Snap a photo, confirm the AI suggestion, tick stages, parts and symptoms	AI suggestion, lists filtered to <i>this plant in this region</i> , legal status
4	Level & stage photos	Enter your sample count or pick a level, then take one photo per stage	% infestation → level from the region's rule, photo quality check, AI count
5	Action, area & send	Control used, yield loss, area, Submit	Weather, vegetation, unit conversion, review summary, offline queue
6	Risk map	Filter and read	Hexagon risk index, trend, alerts, export
7	AI Lab	Photograph a trap or leaf	Identification, counting, % damage, expert queue

CropProtect needed 8 pages and about 30 fields. PestLand collects the same information in 5 screens with about 12 typed or tapped inputs.

5. Click 1 — Where & what type



GPS (automatic). Latitude and longitude are filled in with an accuracy badge:

Badge	Accuracy	What to do
● Good	≤ 10 m	Continue
● Fair	10–30 m	Continue, or step into the open and wait a few seconds
● Poor	> 30 m	PestLand asks you to wait or confirm the areas by hand

Admin areas (automatic). PestLand looks up your GPS point in the offline boundaries of your region pack and fills in **every level**. In Hawai'i that is State > Island > County > District. In Kenya it is County > Sub-county > Ward, the same as CropProtect. Tap any level to change it.

Boundary check. If the areas you pick don't contain your GPS point, a yellow warning appears. In the CropProtect data about 1 in 100 reports from the three main counties sat more than 80 km from their county's centre. The check stops that kind of error at the source.

Place / farm name is optional, for example *Hōlualoa* or a farm name.

What are you reporting? Pick one:

- **Insect / mite:** any animal pest (includes nematodes, snails and slugs in the pack lists)
- **Disease:** fungi, bacteria, viruses, phytoplasma, abiotic look-alikes
- **Weed:** weeds and invasive plants

- **Not sure:** PestLand continues, makes photos compulsory and sends the report to an expert (*Needs ID*)

Tap **Next: plant** (click 1).

6. Click 2 — Host plant

9:41

Host plant

Agricultural · horticultural · ornamental · forest

Click 2 of 5 · Collect

Collect · Analyse

Field crops · Fruit & veg · **Tree crops**

Ornamental & nursery · Forest & native

Turf & pasture

Q Search 1,240 plants in the Hawai'i list

Coffee
Coffea arabica

Macadamia
Macadamia integrifolia

Cacao
Theobroma cacao

'Ōhi'a lehua
Metrosideros polymorpha

Variety: Kona Typica

Planted: Mar 2019

Growth stage: Flowering · **Green berry** · Ripening · Harvest · Dormant / pruned

COFFEE STAGES ONLY

First seen: 20 Sep 2026

Plant age: 7 y 6 m

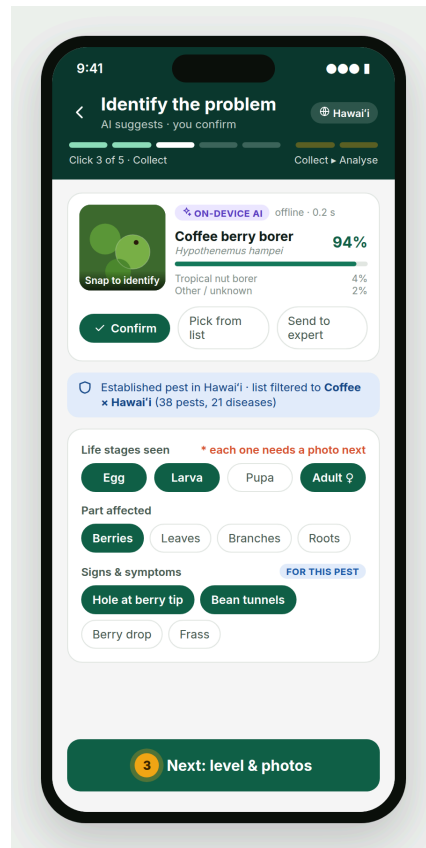
AUTO

2 Next: problem

1. **Sector:** Field crops · Fruit & veg · Tree crops · Ornamental & nursery · Forest & native · Turf & pasture.
2. **Plant:** tap a card, or search the region's list (for example 1,240 plants in Hawai'i). If a plant is missing, type it under **Other**. The report is kept and the region admin is told.
3. **Variety** (optional), e.g. *Kona Typica*.
4. **Growth stage:** only the stages of the plant you picked are shown (coffee: flowering, green berry, ripening, harvest, dormant/pruned).
5. **Planted** and **First seen** dates. PestLand works out the **plant age** and blocks impossible dates, such as planting after the problem was first seen. Choose *Unknown* if the grower doesn't know.

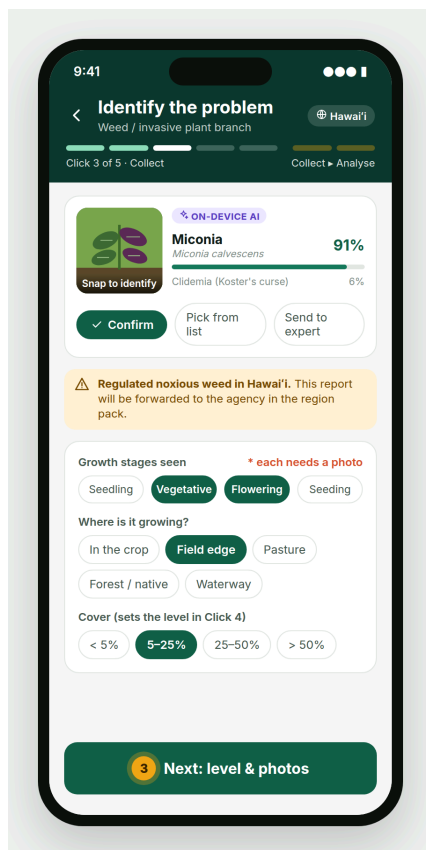
Tap **Next: problem** (click 2).

7. Click 3 — Identify the problem



1. **Snap to identify.** Take one clear photo. The on-device AI (it works offline in about 0.2 s) shows its top three answers with confidence, e.g. *Coffee berry borer* 94 %.
2. Choose one: - **Confirm:** accept the top answer - **Pick from list:** the list shows only problems recorded on **this plant in this region** - **Send to expert:** the report is saved as *Needs ID*
3. **Status badge.** PestLand shows the problem's status in your region: *Established*, *Regulated*, *Quarantine*, *Exotic* or *New to region*. Regulated, quarantine and exotic reports are forwarded to the agency named in the region pack (in Hawai'i, HDOA through the 643-PEST hotline; in Australia, the Exotic Plant Pest Hotline).
4. **Life stages seen:** tick every stage you actually saw (Egg · Larva/Nymph · Pupa · Adult). For diseases the stages are *Early* · *Advancing* · *Late*. **Each ticked stage needs its own photo in Click 4.**
5. **Part affected** and **Signs & symptoms:** multi-select. These lists are filtered to the chosen problem.

7.1 Weed branch



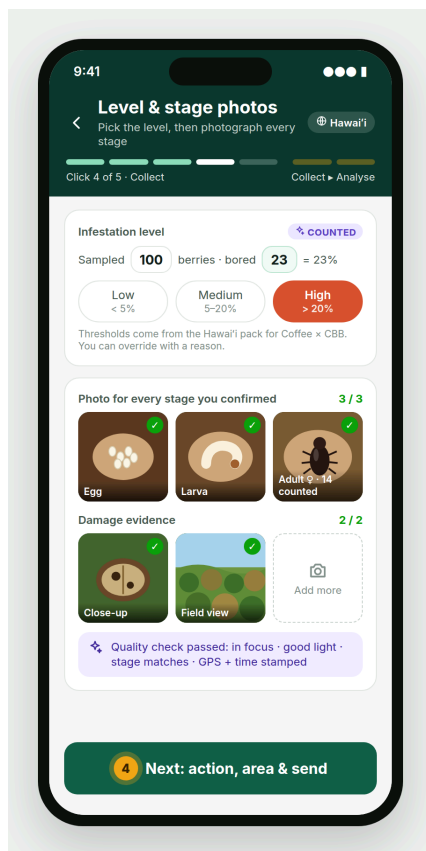
If you picked **Weed** in Click 1, Click 3 shows weed lists on the same screen. CropProtect had a separate weed form.

- **Weed:** confirm the AI answer (e.g. *Miconia*, a regulated noxious weed in Hawai'i), pick from the list or send to an expert.
- **Growth stages seen:** Seedling · Vegetative · Flowering · Seeding. Each ticked stage needs a photo in Click 4.
- **Where is it growing?** In the crop · Field edge · Pasture · Forest/native · Waterway.
- **Cover:** < 5 % · 5–25 % · 25–50 % · > 50 %. This sets the weed's level, so weed reports get a level too. In the CropProtect export all 4,419 weed reports had no level.

The AI is an assistant, not the final word. Your confirmed choice is what is recorded, and the AI's answer is stored alongside it so experts can check both.

Tap **Next: level & photos** (click 3).

8. Click 4 — Level & stage photos



8.1 Select the level first

PestLand replaces CropProtect's bare "Low / Medium / High" with a **count-based level** (the FAMEWS approach):

1. Enter the **sample size** and the number **affected**, e.g. 100 berries sampled, 23 bored.
2. PestLand works out the % (23 %) and picks the level using the region pack's rule for this plant and pest:

Coffee × coffee berry borer, Hawai'i pack (example)	Low	Medium	High
Bored berries per 100	< 5 %	5–20 %	> 20 %

3. If you disagree, tap another level and give a reason. Both values are kept.

If a pest has no counting rule, PestLand uses the pack's default (% plants affected), and you can always pick a level by eye.

8.2 Then photograph every confirmed stage

As soon as the level is set, PestLand opens **one camera slot for every stage you ticked in Click 3**, plus two damage slots:

- **Stage photos:** Egg, Larva, Pupa, Adult (or Early / Advancing / Late for diseases)

- **Damage close-up:** the affected organ
- **Field view:** the wider patch, to show how far the damage has spread
- **Add more:** optional extra photos

The **Next** button stays locked until every ticked stage has a photo that passes the automatic quality check (focus, light, stage visible). Every photo is stamped with GPS and time. When the AI can count (adults on a berry, insects on a trap), the count appears on the photo, e.g. *Adult ♀ · 14 counted*.

8.3 Disease example: one photo per symptom stage

9:41

Level & stage photos

< Disease: photograph each symptom stage Hawai'i

Click 4 of 5 · Collect Collect ► Analyse

Coffee leaf rust *Hemileia vastatrix* · stages ticked in Click 3: Early · Advancing · Late

Infestation level **CALCULATED**

Leaves checked **60** with rust **14** = 23%

Low < 10% **Medium 10-30%** High > 30%

Photo for every symptom stage **3 / 3**

Early · spots Advancing Late · necrosis

Tip: turn the leaf over. Rust spores show on the underside.

Damage evidence **2 / 2**

Close-up Field view Add more

AI leaf-area estimate: 18% of leaf area infected on the sample photos

4 Next: action, area & send

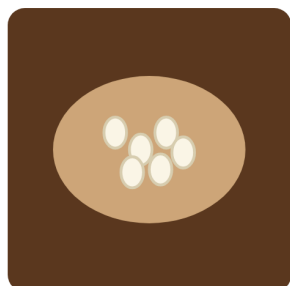
For a disease the stages are **symptom stages**. In this coffee leaf rust example the collector ticked *Early*, *Advancing* and *Late* in Click 3, so Click 4 opens three stage slots. The level comes from leaves

checked and leaves with rust (60 checked, 14 with rust = 23 % = Medium under this example rule). The AI also estimates the % of leaf area infected.

Stage photo guide — one photo per confirmed stage

After choosing the infestation level (Click 4) PestLand opens one camera slot for every stage you ticked in Click 3. A report cannot be sent until each ticked stage has a passing photo.

INSECTS & MITES



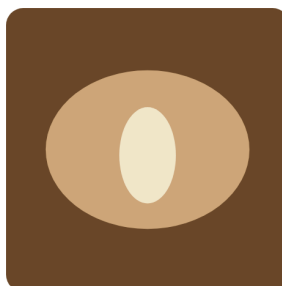
Egg

Macro distance (5–10 cm). Include the leaf/berry surface they sit on.



Larva / nymph

Whole body in frame, side view if possible. Do not squash it.



Pupa

Show the case/cocoon and where it was found (soil, fold, bark).



Adult

Top view, fill the frame. Add a coin or ruler for scale.

DAMAGE, DISEASE & WEEDS



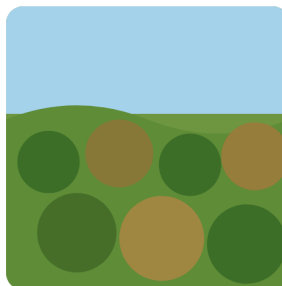
Symptom close-up

The organ with symptoms; hold phone parallel, no shadow.



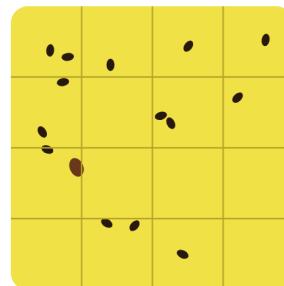
Disease stage

Early (small spot), advancing and late (necrotic) lesions each count as a stage.



Field view

Arms braced, phone at chest height, show the affected patch vs healthy.



Trap / sample card

Whole card flat, even light — AI Lab counts it (Click 7).

Photo tips (from the CropProtect manual, still valid)

- Hold the phone **parallel** to the pest or leaf and don't cast a shadow.
- Get close enough that the pest **fills the frame**. Don't disturb it.
- For the field view, lift the phone to chest height with both hands and brace your elbows against your body.
- Use **Retry** freely. Only accepted photos are kept.
- Photos are compressed for sending (about 350 KB in low-bandwidth packs such as PNG). The original stays on the phone until it has synced.

Tap **Next: action, area & send** (click 4).

9. Click 5 — Action, area & send

Control already done (multi-select): Cultural · Physical/mechanical · Biological · Chemical · None. If you pick Chemical or Biological, choose the **product** from the region's registered list, or type it under *Other*. PestLand records what was done. It does not prescribe treatments.

Context (pre-filled, edit if wrong).

Field	Source	Note
Rain, last 7 days	Weather service (when online) or grower's answer	The source is stored, so observed and reported values stay separate
Vegetation around the field	Suggested from the field-view photo	Green · Greening · Drying · Dry
Estimated yield loss	Grower / you	Limited to 0–100 %

In the CropProtect export, rainfall was blank in 88 % of reports. Filling it automatically fixes that.

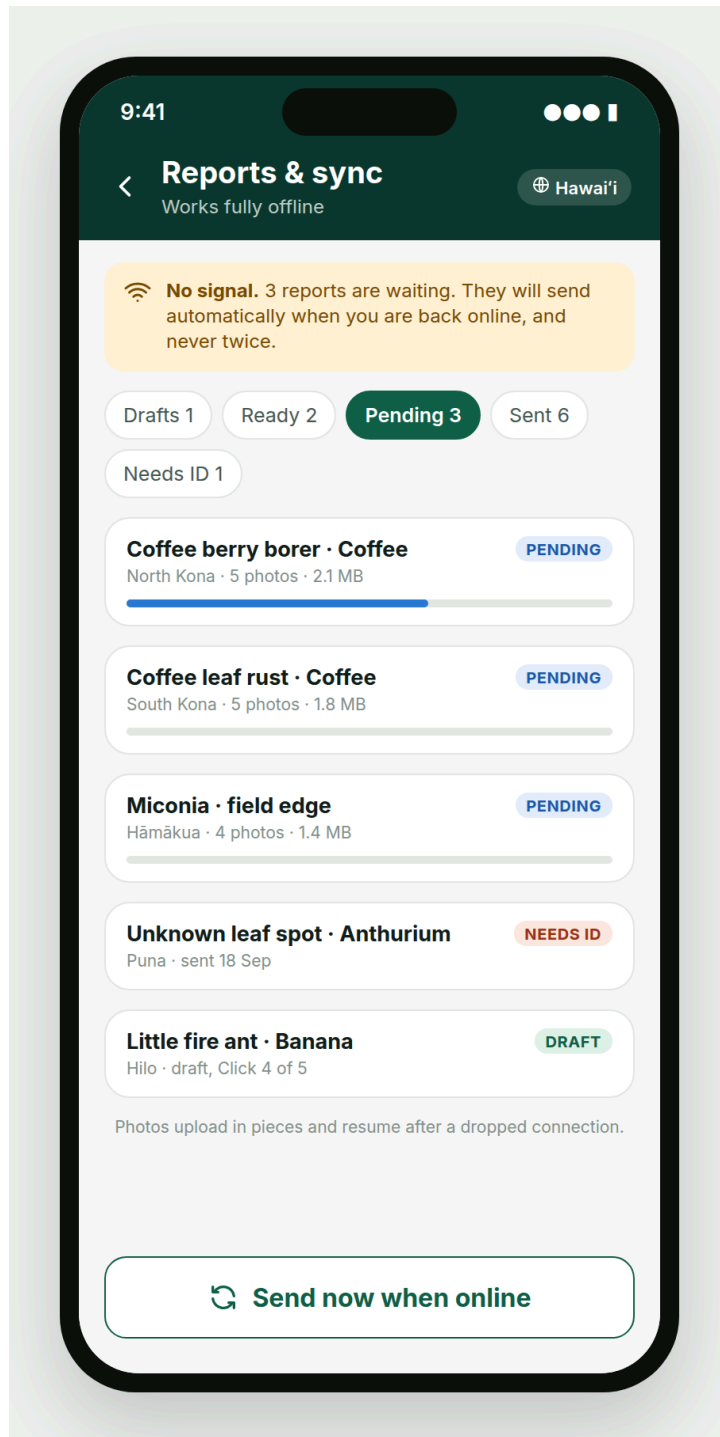
Affected area. Choose one:

- **Type it:** acres or hectares, following the region pack. Both are shown.
- **Walk edge:** walk around the patch and PestLand records the GPS track.
- **Draw:** this is the only place in data collection that opens a map (offline vector tiles). Tap corners, then tap the first corner again to close the shape.

Review. A one-card summary. Tap any line to go back and edit.

Submit (click 5). With a signal the report is sent. Without one it moves to *Pending sync* and goes automatically later.

10. Working offline & sync



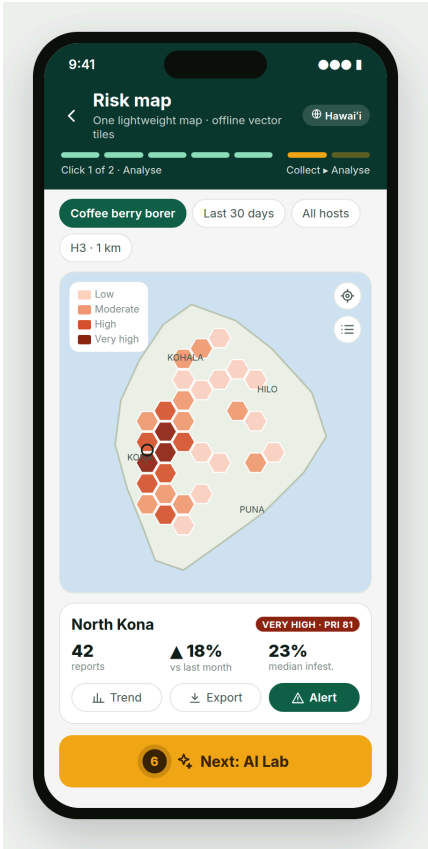
Every report moves through clear states, as in ODK:

Draft → Ready → Pending sync → Sent → Approved / Rejected / Needs ID

- Everything works offline: lists, boundaries, AI, map tiles and the photo checks all live in the region pack on your phone.
- Each report gets a unique ID on the phone, so a retry never creates a duplicate.
- Photos upload in pieces and resume after a dropped connection.

- In packs marked *low connectivity* (e.g. PNG), a short text summary can be sent by SMS and the photos follow later.
- Send the day's reports after fieldwork, as the CropProtect manual advised. PestLand reminds you if anything is still *Pending* at the end of the day.

11. Click 6 — Risk map (results)



The risk map is the **one** analysis map in PestLand. It shows **hexagon cells** (H3, about 1 km across) coloured by the **PestLand Risk Index (PRI)**:

PRI = 100 × (average severity – 1) / 2 × confidence where severity is Low = 1, Medium = 2, High = 3, and confidence = min(1, reports in cell ÷ 5)

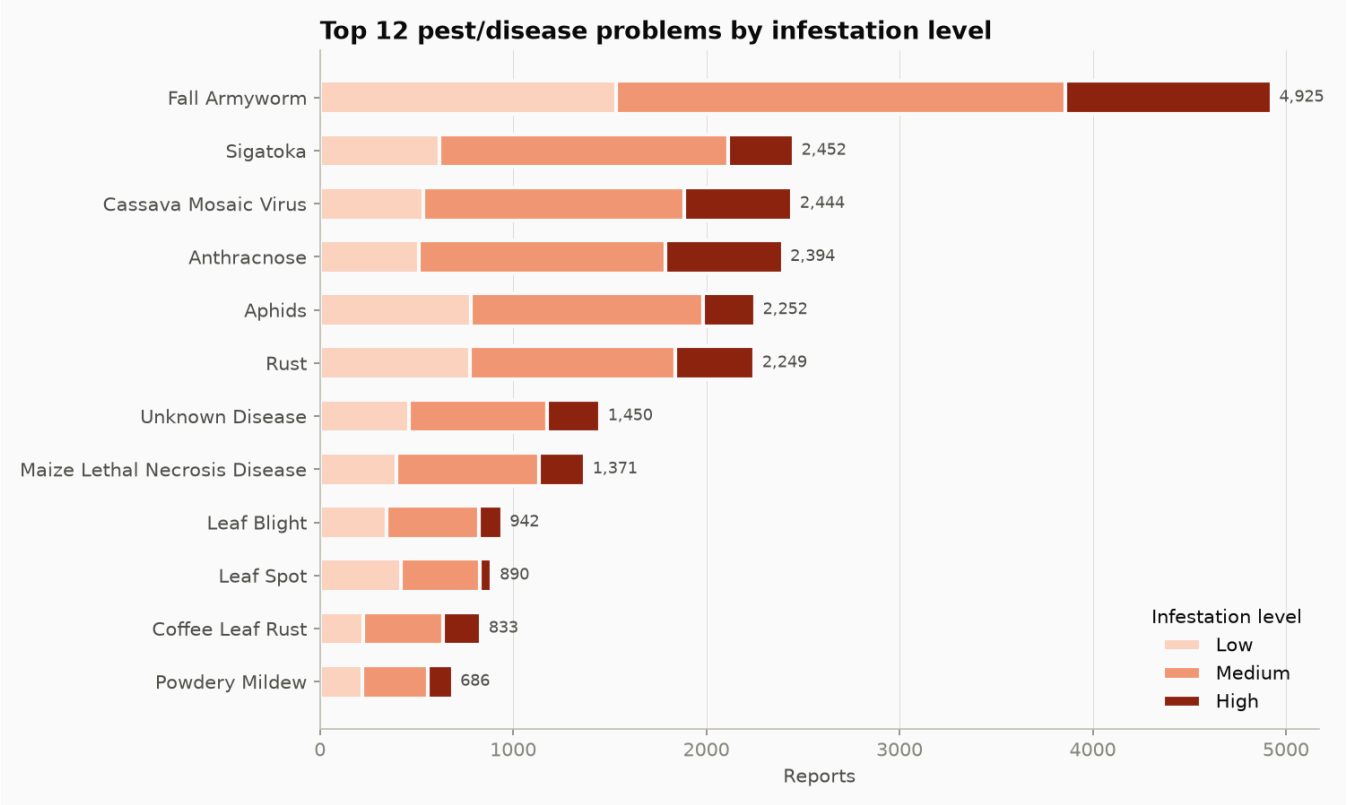
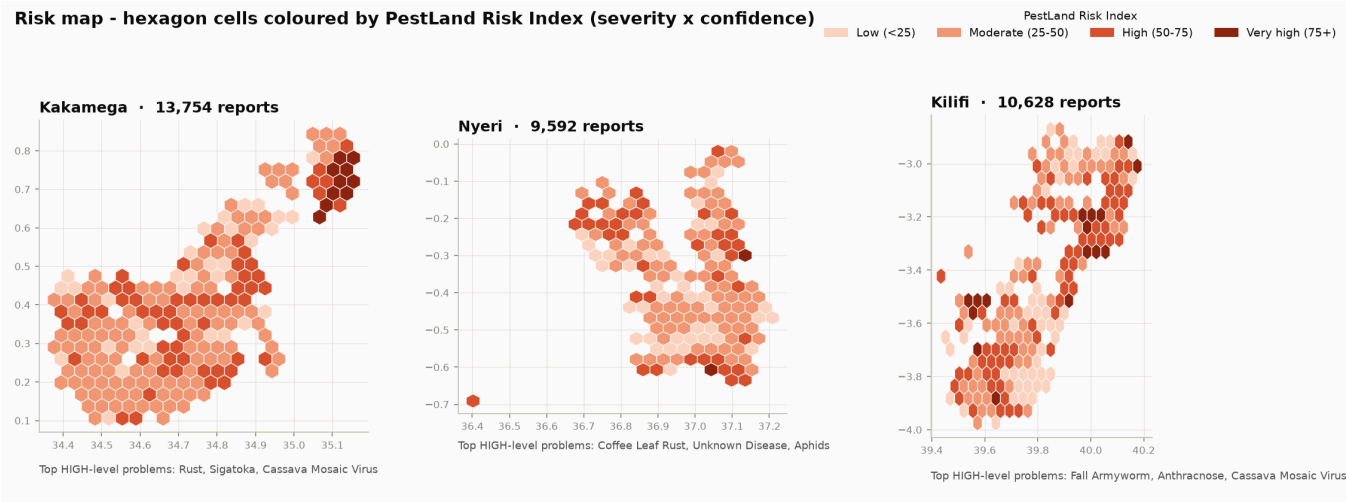
PRI	Class
< 25	Low
25–50	Moderate
50–75	High
≥ 75	Very high

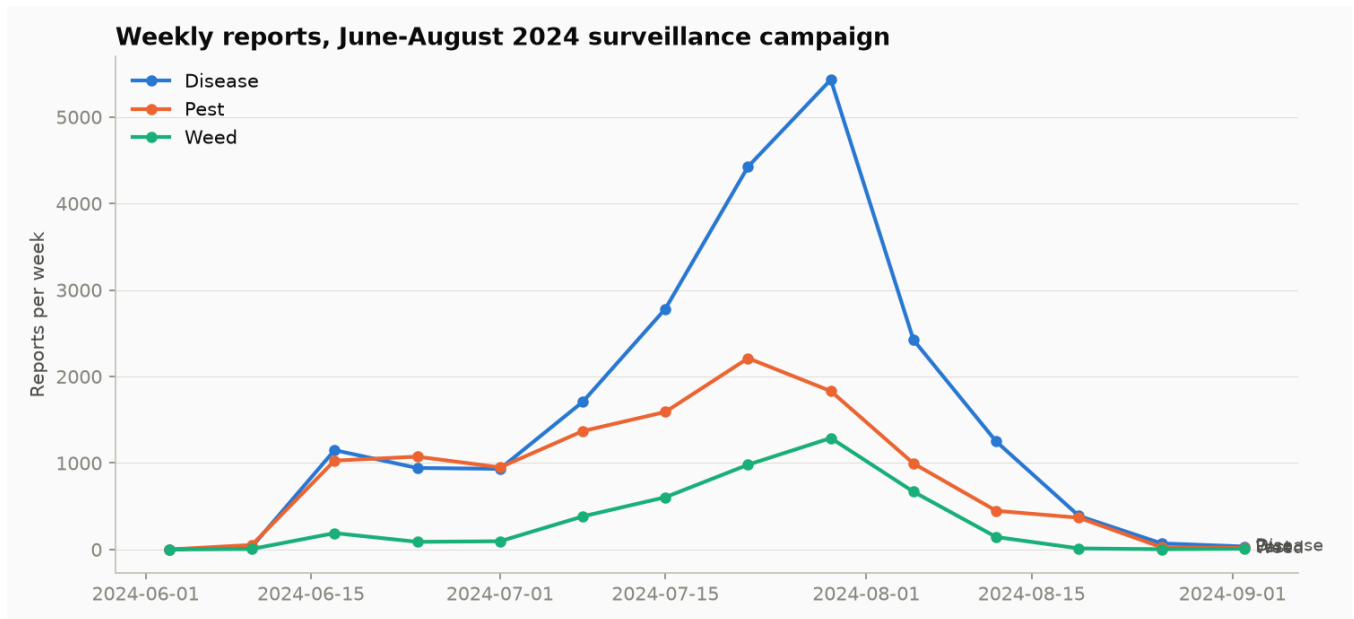
Filters (one row at the top): problem · period · host · cell size. **Tap a cell or area** to see the number of reports, the trend against last month and the median infestation, then use:

- **Trend:** weekly chart
- **Export:** CSV, GeoJSON, Shapefile
- **Alert:** notify the agency or extension officers for that area

11.1 Example from real data (CropProtect export, anonymized)

The figures below come from the 38,762 CropProtect records collected between March 2024 and February 2026: 34,187 pest/disease and 4,547 weed reports. PestLand produces the same views automatically.

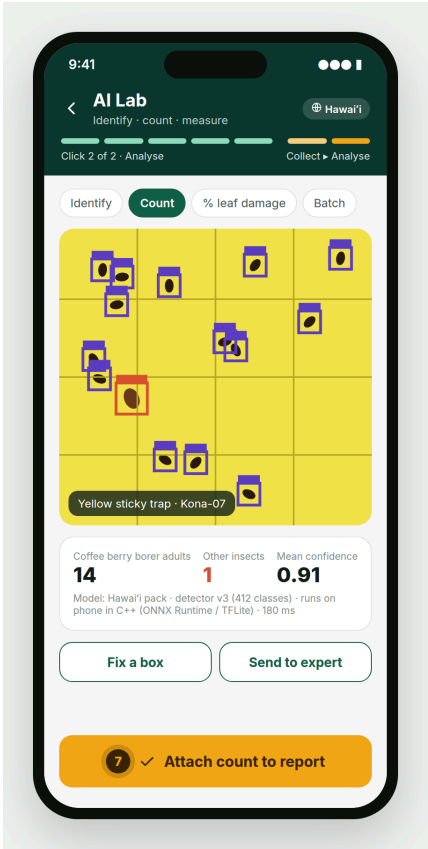




What the data shows, and what PestLand changes because of it:

- **Fall armyworm** was the most reported problem (4,925 pest/disease reports, over 1,000 of them *High*). PestLand's count-based level makes these severities comparable across collectors.
- **1,458 "Unknown disease"** reports had no route to an expert. In PestLand they become *Needs ID* and go into the expert queue with photos of every stage.
- Most reports were collected in a **June–August 2024 campaign**. The weekly trend view makes gaps in surveillance visible.

12. Click 7 — AI Lab (identify & count)

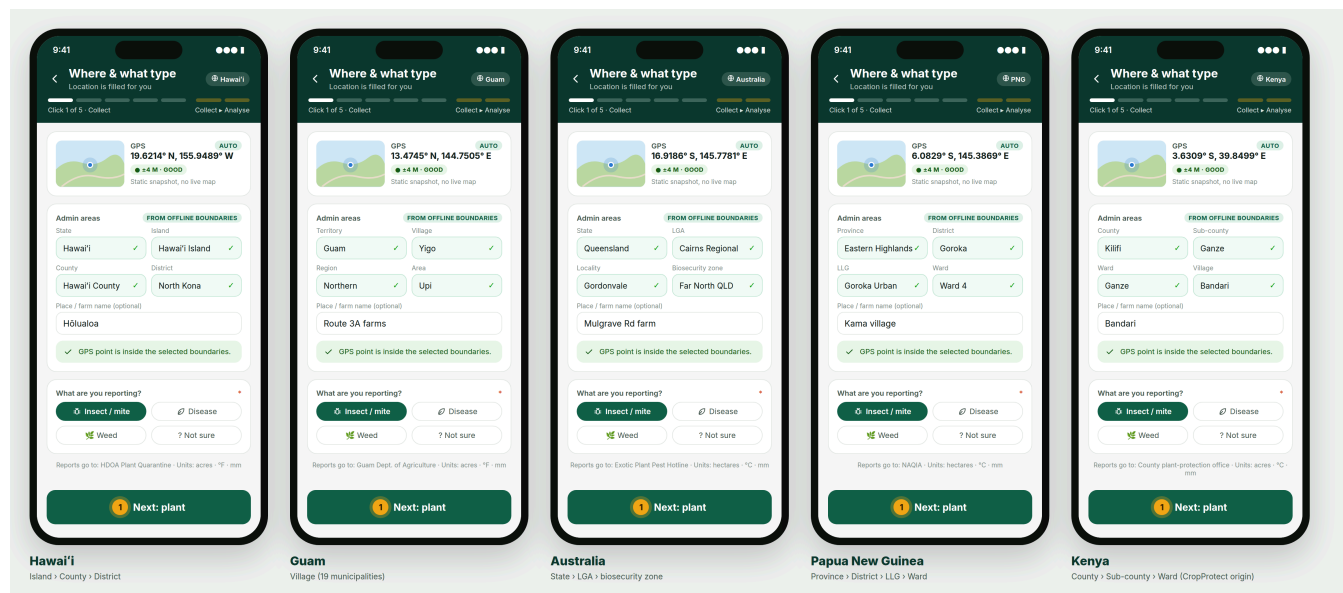


Mode	Use it for	Output
Identify	Any photo of a pest, symptom or weed	Top-3 names with confidence
Count	Sticky traps, pheromone traps, berries, leaves with insects	Boxes around each object + totals per species
% leaf damage	Leaf or canopy photos	% area damaged → suggested level
Batch	A set of trap photos from a route	Counts per trap, sent as trap readings

- The AI runs **on the phone** (native C++ engine) so it works offline. The model comes with the region pack, e.g. *Hawai'i detector v3, 412 classes*.
- **Fix a box**: drag, delete or relabel a box. Your correction is saved and used to improve the next model.
- **Send to expert**: puts the photo in the expert queue.
- **Attach count to report** (click 7) adds the count to the current or a new report.

13. Taking PestLand to another region

"What if I take it to Australia, Guam or PNG?" Nothing in the app changes except the **region pack**. Pick the new region, or let GPS pick it, and every list, rule, map and language switches with it.



What a region pack contains

Selecting a region (or letting GPS pick it) swaps all eight layers at once, so every list, rule and map is confined to that region. New region = new pack, not new code.

Boundaries

Admin levels + names + GeoJSON, e.g. Hawai'i: State > Island > County > District

Host plants

Plants grown/present in the region, grouped by sector, with their growth stages

Pests, diseases, weeds

Filtered per host; status: established - regulated - quarantine - new-to-region

Severity rules

Count/sample methods and Low/Medium/High thresholds per host x pest

Rules & products

Agency to notify, reportable list, registered control products

AI model

Detector trained on that region's species list + label map

Offline map

Clipped PMTiles basemap for the region (the only map data on the phone)

Language & units

Translations, acres/hectares, °F/°C, date format, time zone

Packs shipped at launch: Hawai'i • Guam • USA states • Australia • Papua New Guinea • Kenya (migrated from CropProtect). Packs are signed JSON + GeoJSON + PMTiles + model, versioned like v2026.09.

Region pack	Admin levels (Click 1)	Units	Languages	Example problems	Report channel
Hawai'i	State > Island > County > District	acre, °F	English, 'Ōlelo Hawai'i, Ilocano	Coffee berry borer, coffee leaf rust, coconut rhinoceros beetle, little fire ant, Rapid 'Ōhi'a Death	HDOA Plant Quarantine • 643-PEST
Guam	Territory > Village (19)	acre, °F	English, CHamoru	Coconut rhinoceros beetle, cycad aulacaspis scale, little fire ant	Guam Dept. of Agriculture (set by admin)
USA mainland	State > County (FIPS)	acre, °F	English, Spanish	Spotted lanternfly, emerald ash borer	State Dept. of Agriculture + USDA APHIS

Region pack	Admin levels (Click 1)	Units	Languages	Example problems	Report channel
Australia	State/Territory › LGA › Locality (+ biosecurity zone)	hectare, °C	English	Fall armyworm, Queensland fruit fly, Panama TR4; Xylella (exotic watch)	Exotic Plant Pest Hotline 1800 084 881
Papua New Guinea	Province › District › LLG › Ward	hectare, °C	English, Tok Pisin, Hiri Motu	Cocoa pod borer, coffee berry borer, coconut rhinoceros beetle, fall armyworm	NAQIA (set by admin)
Kenya (from CropProtect)	County › Sub-county › Ward (+ Village)	acre, °C	English, Kiswahili	Fall armyworm, Sigatoka, cassava mosaic, MLN	County plant-protection office

The same pest can have a different status in different regions. Coffee berry borer is *established* in Hawai'i but listed as *regulated* in the PNG example pack, so a PNG report triggers an agency notification and a Hawai'i report does not.

A region that doesn't have a pack yet: a region admin builds one in the web console from boundaries (GeoJSON), plant and pest lists (CSV), severity rules, contacts and translations. PestLand's shared global model is used until a regional AI model has been trained. See [docs/TECHNICAL_SPEC.md §4](#).

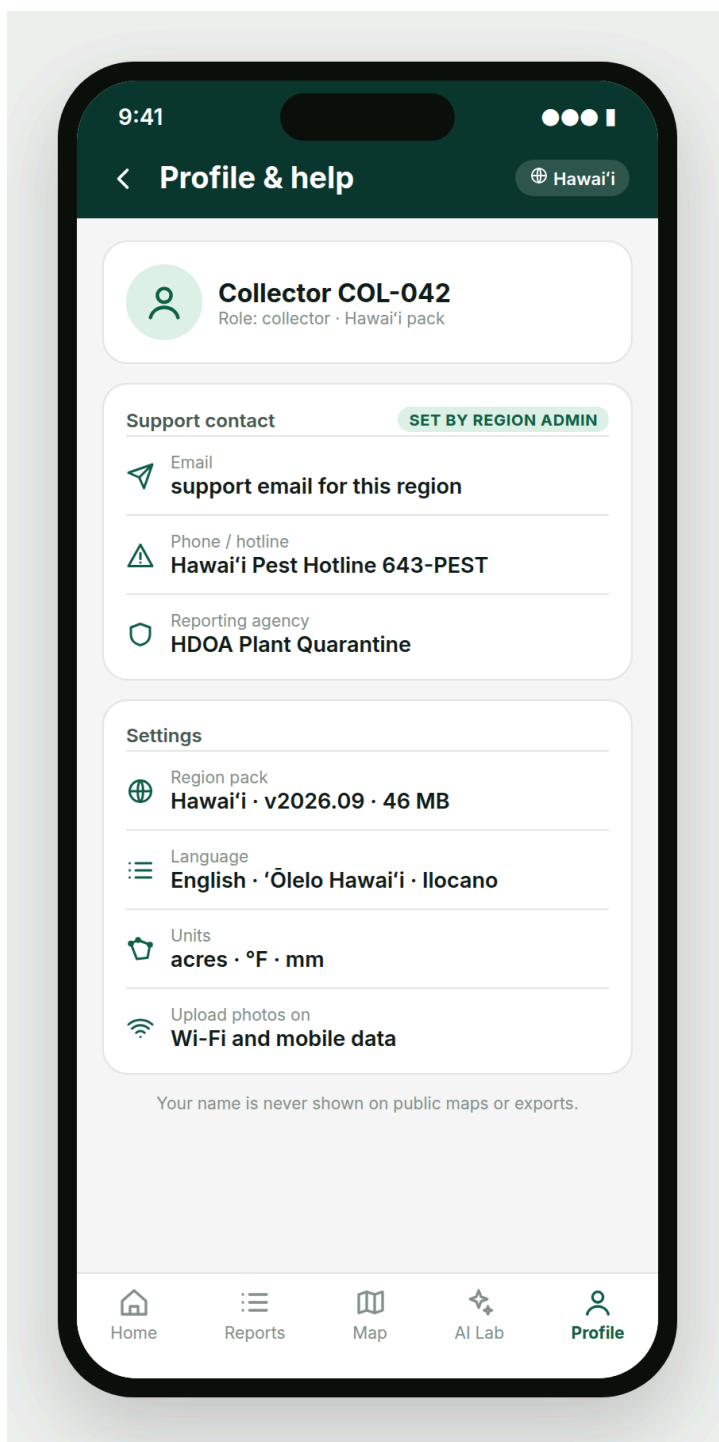
Moving between islands or states: packs can nest (USA › Hawai'i; Australia › Queensland). The app uses the most specific pack that contains your GPS point.

14. CropProtect → PestLand field map

For teams moving from CropProtect. Every CropProtect field is kept or improved.

CropProtect page · field	PestLand click · field	Change
Report type pop-up (Pest/Disease, Weed)	1 · What are you reporting?	Adds <i>Not sure</i> and splits Insect from Disease
Location · Lat/Long	1 · GPS	Adds accuracy badge
Location · County, Sub-county, Ward	1 · Admin areas	Filled automatically. Levels follow the region
Location · Village	1 · Place / farm name	Same, optional
Crop · Crop affected	2 · Sector + Plant	Any plant type, picture cards
Crop · Variety	2 · Variety	Same
Crop · Crop stage	2 · Growth stage	Filtered per plant
Crop · Planting date, Initial detection date	2 · Planted, First seen	Date logic enforced
Crop · Crop age	2 · Plant age	Calculated automatically
Pest · Pest/Disease	3 · AI suggestion + confirm	Adds AI, <i>Needs ID</i> and legal status
Pest · Pest stage	3 · Life stages seen	Multi-select. Each stage needs a photo
Pest · Part affected, Signs/Symptoms	3 · Part affected, Signs & symptoms	Same, filtered
Pest · Level of infestation	4 · Count → level	Count-based, per region rule
Photos · Pest/Disease photo	4 · One photo per stage	1 photo → one per stage
Photos · Damage severity photo	4 · Close-up + Field view	Split in two
Pest · Control measures, Pesticide used	5 · Control done + product	Controlled product list
Weed · Weed, Control, Chemical	3 + 5 (same screens, weed lists)	One flow instead of two
More info · Rainfall, Vegetation	5 · Context	Pre-filled with the source recorded
More info · Yield loss, Additional info	5 · Yield loss, Notes	Limited to 0–100 %
Area · Manual / Mapping / Aerial imagery	5 · Type it / Walk edge / Draw	One offline map, no Google overlay
Submit / Pending submissions	5 · Submit + sync states	Draft › Ready › Pending › Sent
—	6 · Risk map	New
—	7 · AI Lab	New

15. Data, privacy & support



- **Your reports** belong to the programme that runs your region pack. Reviewers in your region see them. The public risk map shows only hexagon summaries, never individual farms.
- **Personal details:** collector names are never published. Exports use pseudonymous collector IDs (e.g. COL-042). The CropProtect sample data in this repository has had collector names replaced this way and GPS rounded to about 100 m. It contained no email or phone columns.
- **Photos** keep their GPS and time for audit and are shared only through signed links.
- **Support:** use **Profile > Help** in the app. It shows the support email, phone or hotline and reporting agency that your region admin has set for your region pack.

Well done. Every report you send makes the risk map sharper for everyone in your region.